Mason Kim

Washington, D.C. Metro Area
sjmasonkim@gmail.com | LinkedIn | Github | Portfolio

Full-Stack Software Engineer with 4+ years of experience building scalable and responsive web applications. Proficient in React, Flask, and SQL with experience developing within secure government environments. Strong focus on clean, maintainable code and collaborative development practices that ensure reliable, production-ready solutions.
Active US Government Security Clearance: Secret.

Technical Expertise

Frontend: HTML5, CSS3, JavaScript, React, Next.js, Vite, Ant Design, Material UI, Bootstrap, Tailwind
Backend & Database: Python, Node.js, Express, Flask, SQLAlchemy, PostgreSQL, MongoDB, REST APIs
DevOps & Cloud: Gitlab CI/CD, DoD Cloud (MIP-SF), Automated Testing, Deployment Pipelines
Security & Compliance: STIG/SRG Compliance, IV&V Processes, WCAG Accessibility, ATO Requirements, Cypress E2E Testing

Professional Experience

JDSAT

Full-Stack Developer | *2021 - Present*

- Designed and developed mission-critical government web applications to streamline data workflows and improve operational efficiency.
- Built and maintained automated testing framework, reducing manual OA by 90% and achieving 99.5% system reliability through CI/CD integration.
- Maintained secure development pipelines in GitLab CI/CD for government cloud environments, reducing deployment errors.
- Conducted STIG compliance reviews and supported IV&V testing, contributing to successful Authority to Operate (ATO) approvals.
- Implemented accessibility enhancements to meet WCAG standards, expanding usability for all users and ensuring full compliance.
- Collaborated with cross-functional teams and stakeholders to define requirements, gather feedback, troubleshoot issues, and deliver production-ready solutions aligned with mission-critical goals.
- Participated in peer code reviews to maintain consistency, ensure quality, and share best practices across the development team.
- Conducted troubleshooting and debugging to identify and resolve issues efficiently, improving system stability and user experience.

Howard Hughes Medical Institute

Research Analyst | *2017 - 2020*

- Executed complex neural analysis and large-scale data processing for groundbreaking Drosophila connectome project, developing innovative analytical approaches to handle massive datasets and extract meaningful biological insights
- Contributed essential dataset creation and computational analysis that directly supported published research in “A Connectome and Analysis of the Adult Drosophila Central Brain”, advancing scientific understanding through rigorous data.

Education & Certifications

George Mason University - *Bachelors of Science, Neuroscience* | *2016*
George Washington University - *Full-Stack Development Certificate* | *2020*
Amazon Web Services - *AWS Certified Cloud Practitioner* | *2024*

Projects

PREP - Portal for Readiness and Emergency Preparedness

- Delivered enterprise emergency management platform enabling Emergency Managers to collect readiness data and track training compliance across 100+ Military Treatment Facilities.
- Consolidated legacy systems into a modern React + Flask platform with automated dashboards, document management, and DoD-cloud security compliance.
- *Deployed in a secure, private DoD cloud environment*

TEEP - Training Exercise & Employment Plan

- Built React application with a dynamic Gantt chart for managing unit assignments, training phases, and resource requirements.
- Improved operational planning by providing real-time visibility into training schedules and unsourced requirements.
- *Live Demo: https://teep.app1.jdsat-labs.com/*

COP - Common Operating Picture

- Developed interactive geospatial mapping application displaying unit deployments, equipment sets, and training phase status.
- Enabled leadership to monitor readiness and deployment planning with dynamic visualization tools.
- *Live Demo: https://cop-poc.app1.jdsat-labs.com/*

PDF Generation Tool

Navy Medicine Enterprise Web Application

- Created an internal tool that converted Excel inputs into standardized PDFs, eliminating manual prep of annual forms.
- Reduced administrative workload, errors, and turnaround time for compliance submissions.
- *Deployed in a secure, private DoD cloud environment*

Amazon Clone

- Built a responsive e-commerce web application featuring advanced UI development, global state management, and modern design principles.
- Implemented shopping cart functionality, dynamic product display, and clean user experience to showcase frontend engineering expertise.
- *Live Demo: https://masonkimm.github.io/amazon-clone/*